

EX NO: 2b	CALCULATION AND VERIFICATION OF FREQUENCY DEVIATION AND MODULATION INDEX OF FM SIGNALS USING MATLAB SIMULATION
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```
%=====
% Calculation and Verification of Frequency Deviation
% and Modulation Index of FM Signal
%=====

clc;
clear;
close all;

%% Message Signal Parameters
Am = 2;           % Message amplitude (V)
fm = 500;         % Message frequency (Hz)

%% Carrier Signal Parameters
Ac = 1;           % Carrier amplitude (V)
fc = 10000;       % Carrier frequency (Hz)

%% FM Parameters
kf = 250;         % Frequency sensitivity (Hz/V)
fs = 100000;      % Sampling frequency (Hz)
T = 0.01;         % Simulation duration (s)

%% Time Vector
t = 0:1/fs:T;

%% Message Signal
m = Am * cos(2*pi*fm*t);

%% Frequency Deviation
delta_f = kf * Am;

%% Modulation Index
beta = delta_f / fm;

%% FM Signal
fm_signal = Ac * cos(2*pi*fc*t + beta*sin(2*pi*fm*t));

%% Display Parameters
fprintf('===== FM PARAMETERS =====\n');
fprintf('Carrier Frequency (fc)   = %.0f Hz\n', fc);
fprintf('Message Frequency (fm)     = %.0f Hz\n', fm);
fprintf('Message Amplitude (Am)     = %.2f V\n', Am);
fprintf('Carrier Amplitude (Ac)     = %.2f V\n', Ac);
fprintf('Frequency Sensitivity (kf)  = %.0f Hz/V\n', kf);

fprintf('\nFrequency Deviation (Δf)   = %.2f Hz\n', delta_f);
fprintf('Modulation Index (β)       = %.2f\n', beta);

%% Verification
beta_verify = delta_f / fm;
```

```
fprintf('Verified Modulation Index = %.2f\n', beta_verify);
```

```
%% Verification Check
```

```
tolerance = 1e-6;
```

```
if abs(beta - beta_verify) < tolerance
```

```
    fprintf('\nVerification Successful\n');
```

```
else
```

```
    fprintf('\nVerification Failed\n');
```

```
end
```

```
%% Frequency Spectrum
```

```
N = length(fm_signal);
```

```
f = linspace(-fs/2, fs/2, N);
```

```
FM_FFT = abs(fftshift(fft(fm_signal))) / N;
```

```
%% Plotting
```

```
figure('Name','FM Modulation','NumberTitle','off');
```

```
% Message Signal
```

```
subplot(3,1,1);
```

```
plot(t, m, 'b', 'LineWidth', 1.5);
```

```
grid on;
```

```
title('Message Signal');
```

```
xlabel('Time (s)');
```

```
ylabel('Amplitude (V)');
```

```
% FM Signal
```

```
subplot(3,1,2);
```

```
plot(t, fm_signal, 'r', 'LineWidth', 1.2);
```

```
grid on;
```

```
title('Frequency Modulated Signal');
```

```
xlabel('Time (s)');
```

```
ylabel('Amplitude (V)');
```

```
% FM Spectrum
```

```
subplot(3,1,3);
```

```
plot(f, FM_FFT, 'k', 'LineWidth', 1.2);
```

```
grid on;
```

```
xlim([fc-3000 fc+3000]);
```

```
title('FM Frequency Spectrum');
```

```
xlabel('Frequency (Hz)');
```

```
ylabel('Magnitude');
```

FM Modulation

FIGURE

FORMAT

TOOLS

LAYOUT

